Git together: the car rental

A play in five acts for two people and two laptops

The story of the Git animation, on your own two laptops, with a car-rental office and GitHub: clone, commit, push and pull, a refused push, a merge, a conflict, and the history that records it all. About 40 minutes.

The cast

Decide who is A and who is B. Sit side by side, each at your own laptop.

A purple scenes: Partner A
B green scenes: Partner B
A B sepia scenes: both

Read your partner's scenes too.

Before you start

- ☐ VS Code and Git work on your laptop (the install party).
- ☐ Each of you has a GitHub account, free at `github.com/signup`.
- ☐ A knows B's GitHub username.
- ☐ The start: `github.com/sef-hs26/git-together`

How to read a scene

1	A step. The same number in the picture shows where.
✓	What you should see. Something else: **Help**, on the last page.
`git push`	At the right end of a step: the Git command behind the click, as the animation's terminal showed it. You never type it.

Five words

commit	A saved version of the project, with your name, the time and a short message.
push	Send your new commits to GitHub, the shared copy.
pull	Get your partner's new commits from GitHub.
merge	Git combines two people's work into one version, in a commit of its own.
conflict	Both of you changed the same line. Git stops, shows both versions, and a person decides.

The program

Two files. The steps name their line numbers.

price.py

```
1  def price(days):
2      amount = days * 60
3      if days >= 7:
4          amount = amount * 0.90    # 10% off
5      return amount
```

The price of a rental: 60 CHF per day, 10% off from 7 days on.

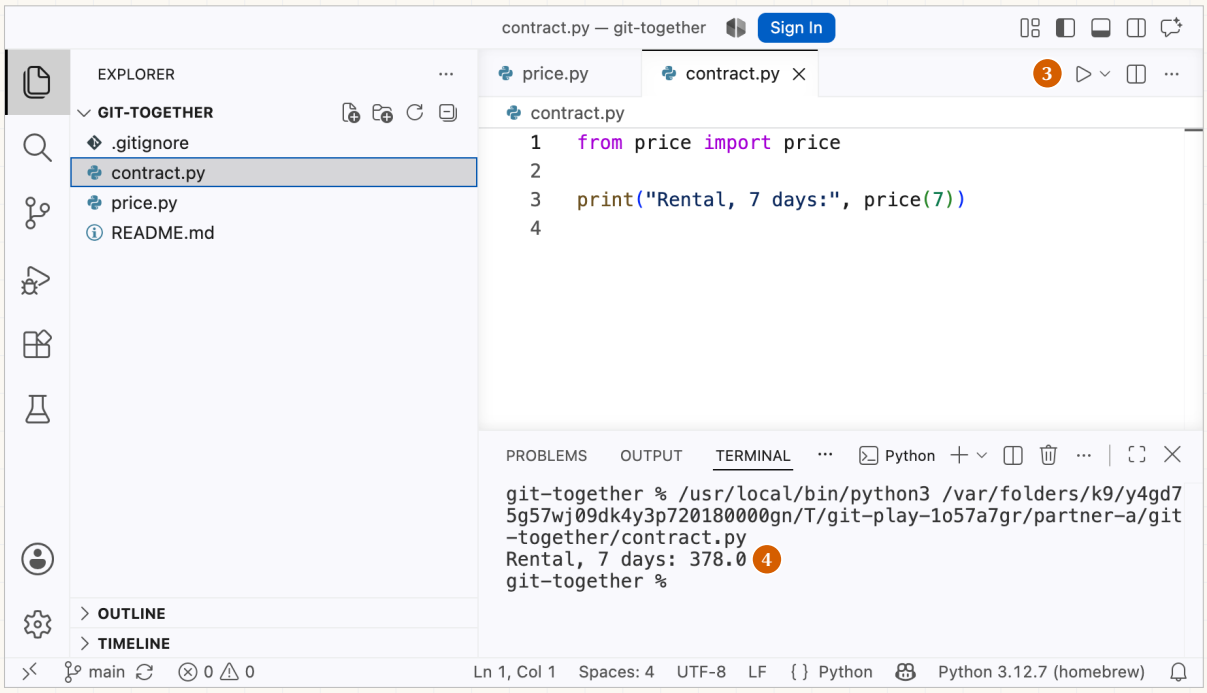
contract.py

```
1  from price import price
2
3  print("Rental, 7 days:", price(7))
```

Prints the price of 7 days: 7 × 60 = 420 CHF, minus 10%: 378.0.

The project also holds `README.md` and `.gitignore`. Leave both as they are.

VS Code

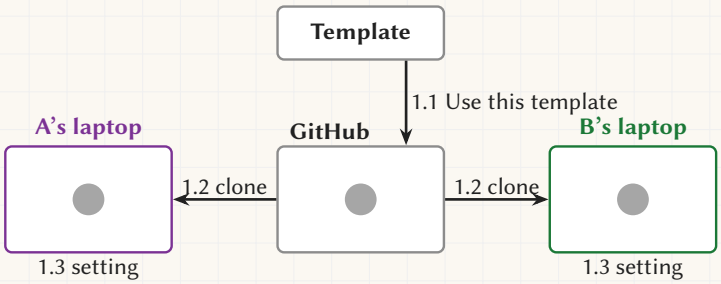


1. **Explorer**: the files of the project.
2. **Source Control**: commit, push, pull, and **GRAPH**, the history.
3. **Run Python File**: runs the open file.
4. The terminal: what the program prints.

ACT 1, ABOUT 10 MINUTES

Setting the stage

A private copy of the project on GitHub for the two of you, and a clone of it on each laptop.



1.1 A repository for the two of you

A

In the browser, on GitHub:

1. Open `https://github.com/sef-hs26/git-together` and sign in.
2. Use this template > **Create a new repository**. Name: `git-together`. **Private**. Create repository.
3. **Settings** > **Collaborators** > Add people: B's username, then **Add … to this repository**.
4. Send B the address of the new repository.
5. B In GitHub's e-mail: View invitation, Accept invitation. No e-mail: open the address with `/invitations` at its end.

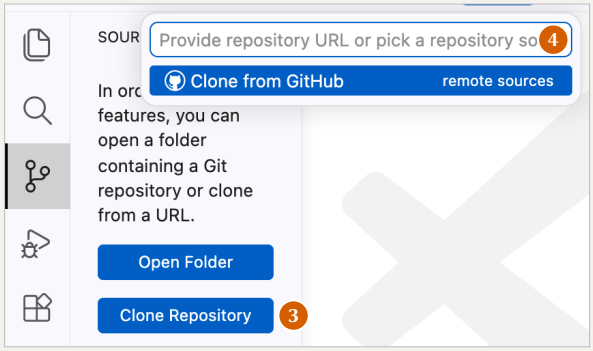
✓ **git-together**, **Private**, four files. B sees the same page.

1.2 Clone

A B

1. GitHub: Code, copy the address that starts with `https://`.
2. VS Code: **File** > **New Window**.
3. **Source Control** > Clone Repository.
4. Paste the address. Enter.
5. A folder, for example **Documents**, then Select as Repository Destination.
6. Open. If asked: Yes, I trust the authors.

`git clone`



✓ Explorer: `.gitignore`, `README.md`, `contract.py`, `price.py`.

✓ **GRAPH**: **Initial commit**.

1.3 One setting

A B

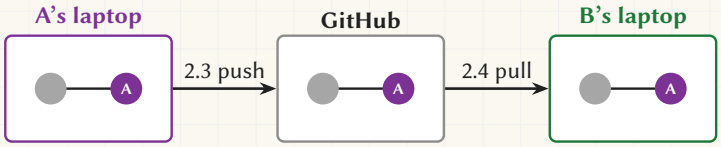
1. **Terminal** > **New Terminal**.
2. `git config pull.rebase false` Enter
3. `git config pull.rebase` Enter

✓ `false`



ACT 2, ABOUT 6 MINUTES

The first change



2.1 change, 2.2 commit

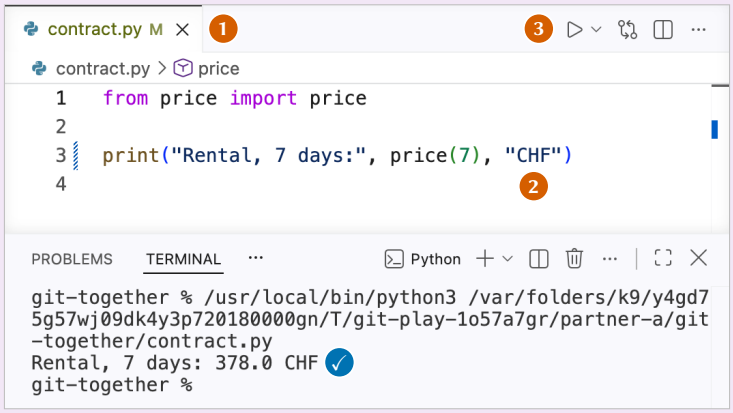
A *A customer phones A: "Your contract says 378.0. But 378.0 what?"*

2.1 Add the currency

A

1. Explorer: click `contract.py`.
2. Line 3: type `, "CHF"` before the last `)`. Save: Cmd+S or Ctrl+S.
3. Click ▷.

✓ `Rental, 7 days: 378.0 CHF`



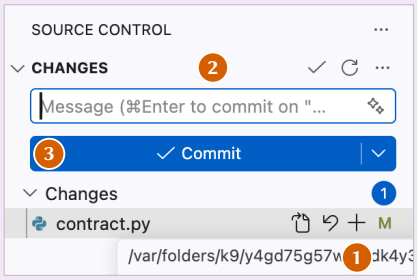
2.2 Commit

A

Source Control:

1. + next to `contract.py`. — `git add`
2. Message: `Show CHF on the contract`
3. Commit — `git commit`

✓ **Changes** is empty.



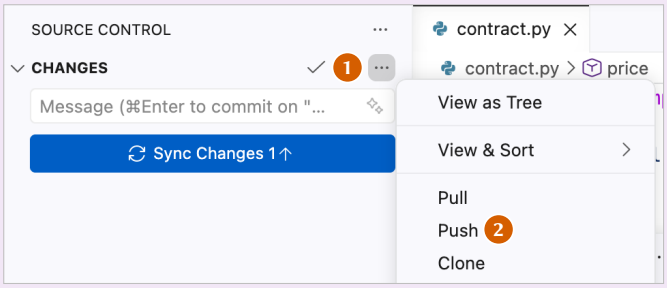
2.3 Push

A

1. ... next to **CHANGES**.
2. **Push** — `git push`

✓ On GitHub (reload the page), the latest commit is **Show CHF on the contract**.

Sync Changes does Pull and Push in one click.



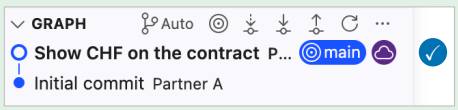
2.4 Pull

B

1. **Source Control**, ..., **Pull**. — `git pull`
2. Run `contract.py` with ▷.

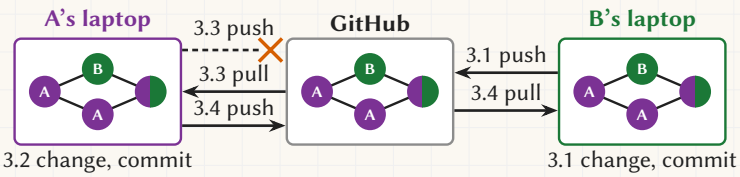
✓ **GRAPH**: **Show CHF on the contract** on top.

✓ `Rental, 7 days: 378.0 CHF`



ACT 3, ABOUT 8 MINUTES

Two changes



B *The fleet manager: "From today, 65 CHF per day."*
A *Marketing: "15% off for a full week."*

3.1 Raise the daily rate

B

1. `price.py`, line 2: `60` → `65`. Save.
2. `+`, `65 CHF per day`, Commit. `git commit`
3. `...` > **Push**. `git push`

✓ GitHub, latest commit: **65 CHF per day**.

price.py M

```
1  def price(days):
2      amount = days * 65
3      if days >= 7:
4          amount = amount * 0.90   # 10% off
5      return amount
```

3.2 Raise the discount, without pulling

A

1. `price.py`, line 4: `0.90` → `0.85` and `10%` → `15%`. Save.
2. `+`, `15% off for a full week`, Commit. `git commit`

✓ Line 2 still says 60: B's change is only on GitHub.

price.py M

```
1  def price(days):
2      amount = days * 60
3      if days >= 7:
4          amount = amount * 0.85   # 15% off
5      return amount
```

3.3 Refused, then Git combines both

A

1. `...` > **Push**: refused, **Can't push refs to remote**. Cancel. `git push`
2. `...` > **Pull**. `git pull`
3. Run `contract.py`.

✓ `price.py`: B's 65 and your 0.85.

✓ `Rental, 7 days: 386.75 CHF`

price.py

```
1  def price(days):
2      amount = days * 65
3      if days >= 7:
4          amount = amount * 0.85   # 15% off
5      return amount
```

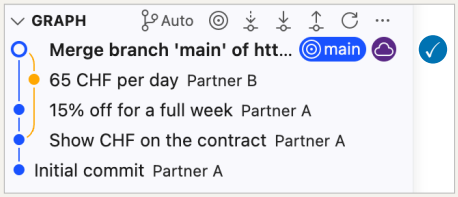
3.4 Share the merge

A B

1. A `...` > **Push**. `git push`
2. B `...` > **Pull**. Run `contract.py`. `git pull`

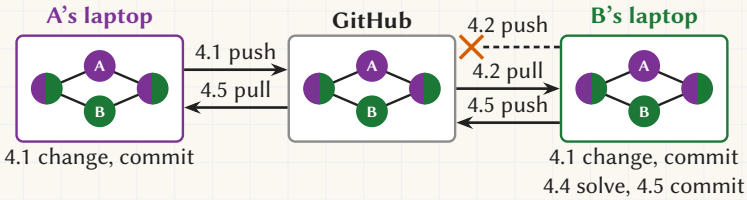
✓ Both **GRAPH**s: **Merge branch ...** on top.

✓ B: `Rental, 7 days: 386.75 CHF`



ACT 4, ABOUT 10 MINUTES

The same line



A *The owner: "Summer season: 70 CHF per day."*

B *The office next door charges 55 CHF. B matches it.*

4.1 Both change line 2, without pulling

A B

1. A `price.py`, line 2: 65 → `70`. Save. `+`, `Summer season: 70 CHF per day`, Commit.
2. B `price.py`, line 2: 65 → `55`. Save. `+`, `Match the competitor: 55 CHF per day`, Commit.
3. A `...` > **Push**. `git push`

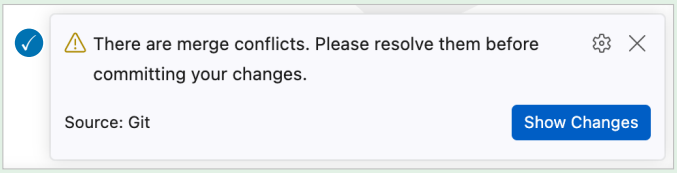
✓ GitHub, latest commit: **Summer season: 70 CHF per day**.

4.2 Refused, then a conflict

B

1. `...` > **Push**: refused. Cancel. `git push`
2. `...` > **Pull**. `git pull`

✓ A warning: merge conflicts.



4.3 Talk and decide

A B

B tells A about the 55 next door, A tells B about the owner's 70. The owner sets the prices: you keep 70.

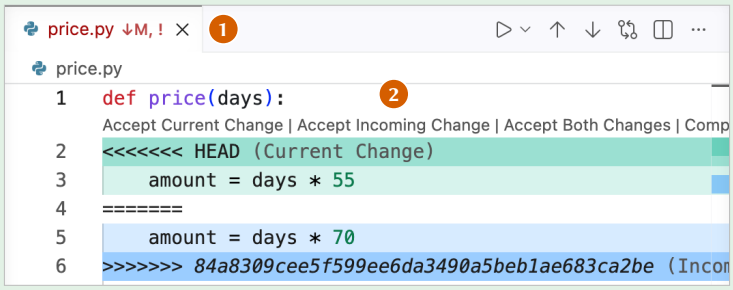
4.4 Solve the conflict

B

1. Open `price.py`.
2. **Accept Incoming Change**: A's 70. Save.
3. Run `contract.py`.

Current Change: yours. **Incoming Change**: from GitHub.

✓ `Rental, 7 days: 416.5 CHF`



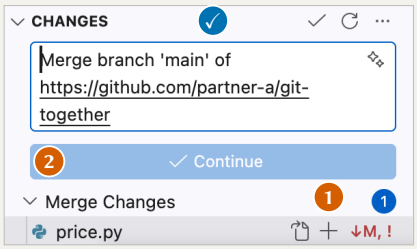
4.5 Commit and share the merge

A B

1. B `+` next to `price.py`. `git add`
2. B Continue. `git commit`
3. B `...` > **Push**. `git push`
4. A `...` > **Pull**. Run `contract.py`. `git pull`

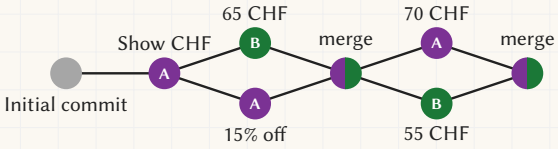
✓ The message, written by Git: **Merge branch ...**

✓ A: `Rental, 7 days: 416.5 CHF`



ACT 5, ABOUT 4 MINUTES

The record



Git kept every step of the play: who, when, and the message that says why.

5.1 Who did what

A B

On GitHub, in the browser:

1. **8 Commits**, above the files.
2. Click **65 CHF per day**.

✓ **Initial commit**, then 4 commits by A and 3 by B. Two start with **Merge branch**.

✓ Line 2 before, in red, and after, in green.

5.2 Why does a car cost 70 CHF?

A B

1. On GitHub: `price.py`, then Blame.

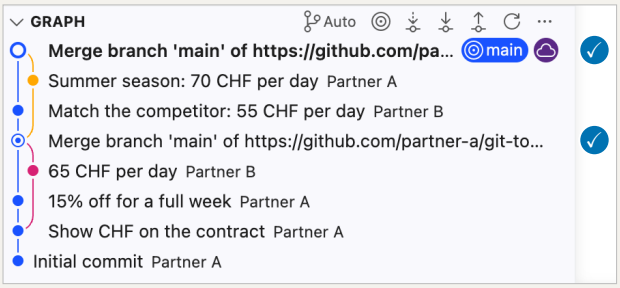
✓ Line 2: **Summer season: 70 CHF per day**, by A. Line 4: **15% off for a full week**, by A.

5.3 The whole picture

A B

1. **Source Control**, **GRAPH**.
2. The mouse over a commit: who made it, and when.

✓ Two merges, from Act 3 and Act 4.



Curtain: three habits

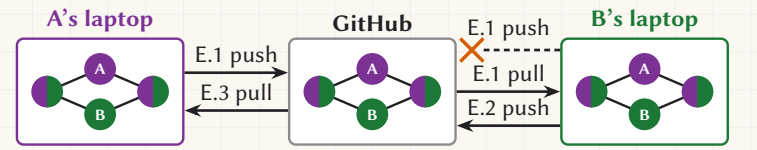
1. **Pull before you push.**
2. **If your push is refused, pull, then push again.**
3. **If you both changed the same line, talk, decide, then + and Continue.**

From now on, **Sync Changes** does the first habit in one click: it pulls, then pushes.

This afternoon, in Task 1, your team works on GitLab, and each of you writes a function in the same file. Git combines your work as in Act 3, as long as you change different lines.

ENCORE, ABOUT 5 MINUTES

Keep both



A *The owner: "Thank our customers on every contract."*

B *The garage: "Remind customers to refuel."*

E.1 Two new last lines

A B

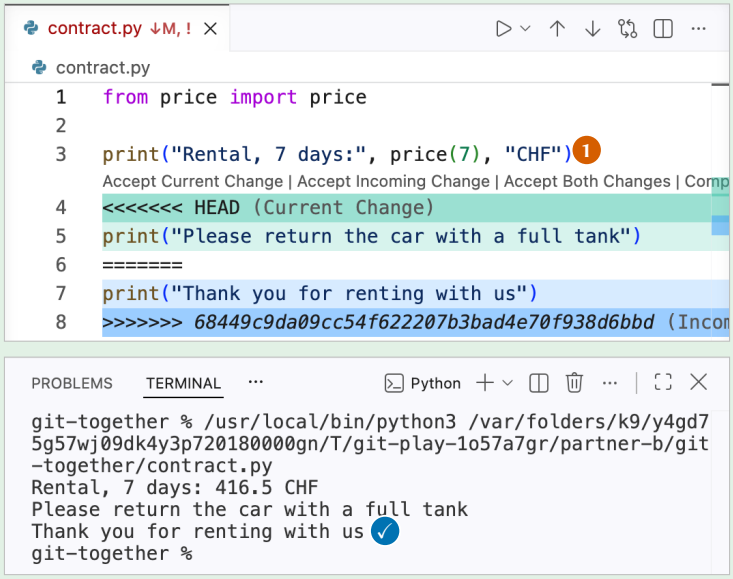
1. A `contract.py`: the end of line 3, Enter, then `print("Thank you for renting with us")`. Save. `+`, message `Thank our customers`, Commit. `...` > **Push**.
2. B Without pulling: the same place, then `print("Please return the car with a full tank")`. Save. `+`, message `Remind customers to refuel`, Commit.
3. B `...` > **Push**: refused, Cancel. `...` > **Pull**: a conflict.

E.2 Keep both

B

1. `contract.py`: **Accept Both Changes**. Save.
2. Run `contract.py`.
3. `+`, Continue. `...` > **Push**.

✓ `Rental, 7 days: 416.5 CHF`, then both new lines.



E.3 Pull

A

1. `...` > **Pull**. Run `contract.py`.

✓ The same three lines as on B's laptop.

Help

Find what you see in the left column. If nothing here fits, call a TA.

You see	Do this
Welcome to VS Code, with an offer to sign in.	Continue without Signing In, or the X at its top right.
None of the registered source control providers work in Restricted Mode.	Manage Workspace Trust, then Trust Folder.
A window asks you to sign in to GitHub.	Mac: Allow. Windows, **Connect to GitHub**: Sign in with your browser. Sign in in the browser, then come back to VS Code.
B sees **Page not found** on A's repository, or a clone says **Repository not found**.	B has not accepted the invitation yet (Scene 1.1, step 5). No invitation: A checks **Settings** > **Collaborators** on GitHub.
GitHub says that **git-together** already exists.	Another name, for example `git-together-2`.
A pull stops with **Need to specify how to reconcile divergent branches**.	Scene 1.3 was skipped. Do it now, then pull again.

Make sure you configure your "user.name" and "user.email" in git.	In the terminal, with your name and the e-mail of your GitHub account, then commit again:
	`git config --global user.name "First Last"`
	`git config --global user.email "first.last@ethz.ch"`

Nothing under **Changes** after an edit.	Save: Cmd+S or Ctrl+S. A dot on the tab of a file means: not saved.
There are no staged changes to commit. Would you like to stage all your changes and commit them directly?	The + was left out. Yes; next time, + first.
Can't push refs to remote. Try running "Pull" first to integrate your changes.	GitHub holds a commit that your laptop lacks. Cancel, ... > **Pull**, then push again.
Sync Changes clicked in place of **Push**.	No harm done: it pulled, then pushed. Do the checks of the next pull scene and go on.
Are you sure you want to stage price.py with merge conflicts?	No. The file still holds marker lines (`<<<<<<<`, `=======`, `>>>>>>>`): click an **Accept** link above them, save, then +.
Continue stays grey.	**Merge Changes** still lists a file: click its +.
A button **Resolve in Merge Editor**.	Leave it. The **Accept** links above the conflict do the job.
No ▷ at the top right of the editor, or Python is not found.	Call a TA. Meanwhile, in the terminal: `python3 contract.py` (Mac) or `python contract.py` (Windows).
A terminal full of lines that start with ~.	That is the editor Vim. Type `:wq` and press Enter.
Your history differs from the act's diagram.	Compare with your partner's **GRAPH**, then call a TA. If all else fails: delete your project folder and clone again (Scene 1.2). What you pushed is safe on GitHub.